

Chapter 6.

6.1. We need to see whether we can find a probability vector (P_1, P_2, P_3) for which all bets are fair. Therefore, it requires

$$O_i P_i - (1 - P_i) = 0, \text{ for } i=1, 2, 3$$

$$\Rightarrow P_i = \frac{1}{1+O_i}$$

From $O_1 = 1, O_2 = 2, O_3 = 5$, we get

$$P_1 = \frac{1}{2}, P_2 = \frac{1}{3}, P_3 = \frac{1}{6}$$

and $P_1 + P_2 + P_3 = \frac{1}{2} + \frac{1}{3} + \frac{1}{6} = 1$. So

$(\frac{1}{2}, \frac{1}{3}, \frac{1}{6})$ is a probability vector making all bets fair. Thus, no arbitrage is present.

6.2. O_4 must satisfy

$$\frac{1}{1+2} + \frac{1}{1+3} + \frac{1}{1+4} + \frac{1}{1+O_4} = 1$$

$$\frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{1+O_4} = 1$$

$$\frac{1}{1+O_4} = 1 - \frac{47}{60}$$

$$\frac{1}{1+O_4} = \frac{13}{60}$$

$$1+O_4 = \frac{60}{13}$$

$$O_4 = \frac{60}{13} - 1 = \frac{47}{13}$$

6.3 (a) If an arbitrage is impossible, then there is a probability vector (P_1, P_2, P_3)

such that

$$\begin{cases} \sum_{j=1}^3 P_j Y_1(j) = 0 \\ \sum_{j=1}^3 P_j Y_2(j) = 0 \\ \sum_{j=1}^3 P_j = 1 \end{cases} \Rightarrow \begin{cases} 4P_1 + 8P_2 - 10P_3 = 0 \\ 6P_1 + 12P_2 - 16P_3 = 0 \\ P_1 + P_2 + P_3 = 1 \Rightarrow P_3 = 1 - P_1 - P_2 \end{cases}$$

$$\Rightarrow \begin{cases} 14P_1 + 18P_2 = 10 \\ 22P_1 + 28P_2 = 16 \end{cases} \Rightarrow \begin{cases} P_1 = 2 \\ P_2 = -1 < 0 \end{cases}$$

impossible!
since $P_2 < 0$ as a probability.

Thus an arbitrage is possible.

(b). Solve a probability vector (P_1, P_2, P_3) such that

$$\begin{cases} 6P_1 - 3P_2 + 0P_3 = 0 - \textcircled{1} \\ -2P_1 + 0P_2 + 6P_3 = 0 - \textcircled{2} \\ 10P_1 + 10P_2 + xP_3 = 0 - \textcircled{3} \\ P_1 + P_2 + P_3 = 1 - \textcircled{4} \Rightarrow P_3 = 1 - P_1 - P_2 \end{cases}$$

$$\Rightarrow \begin{cases} 6P_1 - 3P_2 = 0 \\ -8P_1 - 6P_2 = -6 \end{cases} \Rightarrow \begin{cases} P_1 = \frac{3}{10} \\ P_2 = \frac{3}{5} \end{cases} \Rightarrow P_3 = \frac{1}{10}$$

It then gives us

$$10 \cdot \frac{3}{10} + 10 \cdot \frac{3}{5} + x \cdot \frac{1}{10} = 0$$

$$x = -90$$

6.4.

outcome	odds
1 or 2	O_{12}
2 or 3	O_{23}
3 or 1	O_{31}

← new odds except those in 6.1.

If no arbitrage is present, then the probability vector $(P_1, P_2, P_3) = (\frac{1}{2}, \frac{1}{3}, \frac{1}{6})$ also need satisfy

$$\begin{cases} \sigma_{12}(P_1 + P_2) - P_3 = 0 \\ \sigma_{23}(P_2 + P_3) - P_1 = 0 \\ \sigma_{31}(P_3 + P_1) - P_2 = 0 \end{cases}$$

$$\Rightarrow \sigma_{12} = \frac{P_3}{P_1 + P_2} = \frac{1/6}{1/2 + 1/3} = \frac{1}{5}$$

$$\sigma_{23} = \frac{P_1}{P_2 + P_3} = \frac{1/2}{1/3 + 1/6} = 1$$

$$\sigma_{31} = \frac{P_2}{P_3 + P_1} = \frac{1/3}{1/6 + 1/2} = \frac{1}{2}$$

6.5. Suppose the outcome is j , we get the gain

$$\begin{aligned} O_j X_j - \sum_{i \neq j} X_i &= O_j X_j - \left(\sum_{i=1}^m X_i \right) + X_j \\ &= (1+O_j) X_j - \sum_{i=1}^m X_i \\ &= (1+O_j) \frac{(1+O_j)^{-1}}{1 - \sum_{i=1}^m (1+O_i)^{-1}} - \sum_{i=1}^m \frac{(1+O_i)^{-1}}{1 - \sum_{i=1}^m (1+O_i)^{-1}} \\ &= \frac{1 - \sum_{i=1}^m (1+O_i)^{-1}}{1 - \sum_{i=1}^m (1+O_i)^{-1}} = 1 \end{aligned}$$

6.6. From the Example 6.1b, we have

$$P_1 = P = \frac{1+2r}{3}, \quad P_2 = 1-P = \frac{2-2r}{3} = \frac{2(1-r)}{3}$$

The present value return from purchasing one put option is

$$\text{return} = \begin{cases} -P & \text{if the price is 200 at time 1.} \\ 100(1+r)^{-1} - P & \text{if the price is 50 at time 1.} \end{cases}$$

⇒ In order for there to be no arbitrage, we have

$$E[\text{return}] = P_1 \cdot (-P) + P_2 [100(1+r)^{-1} - P] = 0$$

$$\frac{1+2r}{3} \cdot (-P) + \frac{2(1-r)}{3} [100(1+r)^{-1} - P] = 0$$

$$\frac{2(1-r)}{3} \cdot \frac{100}{1+r} - P = 0$$

$$P = \frac{200(1-r)}{3(1+r)} \quad \text{it is easy to check } S + P - C = K(1+r)^{-1}$$

6.7. We need find $C(K, t)$ where $K=150, t=2$.

$$S=100, u=2, d=1/2$$

$$\Rightarrow S_{uu}=400, S_{ud}=S_{du}=100, S_{dd}=25$$

⇒ The corresponding payoffs are.

$$C_{uu} = (400-150)(1+r)^{-2} = 250(1+r)^{-2}$$

$$C_{ud} = C_{du} = 0$$

$$C_{dd} = 0$$

Let P denote the risk neutral probability that the price of the stock goes up. We then know

$$P = \frac{1+r-d}{u-d} = \frac{1/2+r}{3/2} = \frac{1+2r}{3}$$

thus

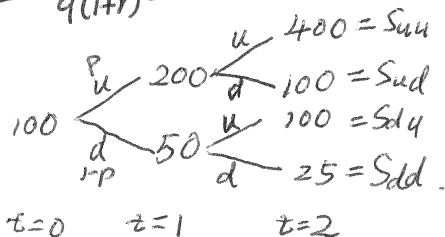
$$C = P_{uu} \cdot C_{uu} + P_{ud} \cdot C_{ud} + P_{du} \cdot C_{du} + P_{dd} \cdot C_{dd}$$

$$= P_{uu} \cdot C_{uu}$$

$$= P^2 \cdot C_{uu}$$

$$= \left(\frac{1+2r}{3}\right)^2 \cdot 250(1+r)^{-2}$$

$$= \frac{250(1+2r)^2}{9(1+r)^2}$$



6.8. By the arbitrage theorem, there

will be no sure win if there is a probability vector $(P_1, P_2, 1-P_1-P_2)$ on the outcomes that make the expected return equal to zero for both wagers.

For purchasing one share of the stock,

$$\text{return} = \begin{cases} 200(1+r)^{-1} - 100 & \text{if the price is 200 at time 1} \\ 100(1+r)^{-1} - 100 & \text{if the price is 100} \\ 50(1+r)^{-1} - 100 & \text{if the price is 50} \end{cases}$$

$$E[\text{return}] = P_1 \left(\frac{200}{1+r} - 100 \right) + P_2 \left(\frac{100}{1+r} - 100 \right) + (1-P_1-P_2) \left(\frac{50}{1+r} - 100 \right) = 0$$

$$\Rightarrow 3P_1 + P_2 = 1 + 2r \quad \text{--- (1)}$$

For purchasing one option,

$$\text{return} = \begin{cases} 50(1+r)^{-1} - C & \text{if the price is 200} \\ -C & \text{if the price is 100} \\ -C & \text{if the price is 50.} \end{cases}$$

$$E[\text{return}] = P_1 \cdot \left(\frac{50}{1+r} - C \right) + P_2 \cdot (-C) + (1-P_1-P_2) \cdot (-C) = 0$$

$$\Rightarrow C = \frac{50P_1}{1+r} \quad \text{--- (2)}$$

From (1), we have

$$3P_1 = 1 + 2r - P_2$$

$$P_1 = \frac{1+2r-P_2}{3}$$

Since $0 \leq P_2 \leq 1$, we get

$$\frac{2r}{3} \leq P_1 \leq \frac{1+2r}{3}$$

$$\text{thus } \frac{100r}{3(1+r)} \leq C = \frac{50P_1}{1+r} \leq \frac{50(1+2r)}{3(1+r)}$$